

Ambient Expense Approval Agent

Business Helper for Automated Expense Tracking & Authorization

What It Does

This project is an ambient business helper that automates expense ingestion and tracking. It processes receipt images, audio recordings, or raw text, extracts transaction metadata, and handles normalization, policy checking, and ledger logging to streamline corporate bookkeeping.

Key Added Features

- **Currency Normalization Skill:** Normalizes raw amount strings (e.g., ₹500, Rs. 500/-) to standardized amounts and ISO currency codes.
- **Reconciliation Agent:** Integrates lookup checks against the SQLite ledger database to flag duplicates and resolve vendor/party ambiguities.
- **PII & Injection Security Checkpoint:** Scrubs sensitive PII data and intercepts prompt-injection payloads on all transaction values early.
- **HITL Dashboard Console:** Real-time manual review interface showcasing decision logs, flagged reasons, and original submission text for bookkeepers.

Agentic Graph Workflow

START → *Intake Agent* (extracts metadata) → *Reconciliation Agent* → **[If Ambiguous]** → *HITL Console* → **Record Outcome**
[If Clean] → *Security Checkpoint* (PII / Injection Check) → *Parse Expense Event* → **[If < \$100]** → *Auto Approve* → **Record Outcome**
[If ≥ \$100] → *LLM Risk Review* → *Human Approval* → **Record Outcome** (writes to SQLite ledger and log).

Technology Stack

Core Framework	Google Agent Development Kit (ADK) 2.0 Graph Workflow
Language & Server	Python, FastAPI, Uvicorn, SQLite
AI Models	Gemini 3.1 Flash Lite (GenAI SDK)
Testing & Quality	pytest, pytest-asyncio, ruff linter, HTTP mock triggers